Team ID: 23 Megathon X: Singularity — Milestone 1 (M1)

Automatic Category Suggestion for Wikimedia Commons via Cascaded Dense Retrieval and Hierarchical Graph Verification

Abstract—Wikimedia Commons categorizes millions of uploads across an open-ended, heavy-tailed Directed Acyclic Graph (DAG) plagued by unlabelled ancestors, cyclic links, and missing labels[cite: 3]. We propose an end-to-end, sub-second cascaded retrieve-and-rerank architecture[cite: 4, 5]. Stage 1 generates high-recall visual candidates across millions of categories in < 30 ms using a frozen SigLIP backbone coupled with FAISS-based k-NN image retrieval, zero-shot text prototypes, and an Asymmetric Loss (ASL) linear head[cite: 4, 5]. Stage 2 proactively discovers events from wikitext and EXIF metadata, resolves temporal/spatial contradictions, and enforces Commons' "most-specific" cataloging rule via topological ancestor suppression[cite: 3, 4]. Stage 3 resolves fine-grained visual ambiguities using a single-turn, grammar-constrained Vision-Language Model (VLM) reranker that outputs calibrated predictions alongside human-auditable rationales[cite: 4, 5]. The entire pipeline operates in < 170 ms, scales via append-only vector updates without retraining, and evaluates on a temporally deduplicated split[cite: 3, 4, 5].

1 Problem Statement & Understanding

Let $\mathcal{C}$ represent the open set of Commons categories ($|\mathcal{C}| \gg 10^6$), structured as a directed graph $G = (\mathcal{C}, E)$[cite: 3]. Given an uploaded image x and optional metadata m (title, description, EXIF timestamp, geotags), the objective is to produce a ranked list of suggestions $\hat{Y}_k(x, m) = \text{top-}k_{c \in \mathcal{C}} s(c \mid x, m)$ matching volunteer editor annotations $Y(x)$[cite: 3]. This presents distinct engineering challenges:

- **Extreme Label Space & Asymmetric Absence:** Most categories reside in an extreme long tail, while volunteer labeling is incomplete[cite: 3]. Unassigned categories cannot be treated as true negatives, and missing parent categories are systematically unlabelled because editors tag only the most specific leaves[cite: 3, 4].
- **Heterogeneous Semantic Modalities:** Visual attributes identify objects and scenes, but events, historical context, and municipal jurisdictions require joint temporal, spatial, and linguistic reasoning[cite: 3].
- **Graph Irregularities:** While defined as a DAG in principle, real Commons category structures contain noisy cycles that disrupt naive tree search[cite: 3].
- **Throughput & Continual Learning Constraints:** With 10^4–10^5 daily uploads (~ 1 image/s), inference must be bounded-cost (< 1 s)[cite: 3, 5]. Newly registered categories and uploads must be integrated continuously without retraining backbones[cite: 3, 4].

2 Proposed Pipeline Architecture

We decompose category assignment into a three-stage pipeline that cascades from broad high-throughput vector retrieval down to fine-grained semantic verification[cite: 4, 5].

High-Level Inference Pipeline Flow

Upload $(x, m) \longrightarrow$ **SigLIP Vision Encoder** (z_x)[cite: 4]

► **Stage 1: Multi-Scorer Candidate Recall** (< 30 ms)[cite: 4, 5]
- Scorer 1: FAISS IVF-PQ k-NN consensus voting (S_{knn})[cite: 4, 5]
- Scorer 2: Zero-shot text-prototype cosine match (s_{txt})[cite: 4]
- Scorer 3: ASL supervised head on head classes (p_c)[cite: 4]

$\Longrightarrow$ Candidate Pool C_{visual} = Top-50 candidates[cite: 4]

► **Stage 2: Metadata Event Discovery & DAG Pruning** (< 60 ms)[cite: 4, 5]
- Parse EXIF year (Y), reverse-geocode GPS coordinates offline[cite: 3]
- Proactive LLM event formulation $\rightarrow$ Query event index[cite: 3, 4]
- Prune chronological/geographic contradictions[cite: 3]
- Category DAG ancestor suppression (enforce leaf specificity)[cite: 3, 4]

$\Longrightarrow$ Filtered Candidate Pool C_{stage2} (10–15 categories)[cite: 4]

► **Stage 3: VLM Grounding & Reranking** (< 60 ms)[cite: 4, 5]
- Single-turn grammar-constrained pass (Florence-2 / PaliGemma)[cite: 4, 5]
- Visual attribute verification & sibling leaf disambiguation[cite: 4]
- Score fusion: $s_{\text{final}} = \alpha \cdot s_{\text{prior}} + \beta \cdot s_{\text{VLM}}$[cite: 4]

$\Longrightarrow$ **Ranked Suggestions** $\hat{Y}_k(x, m)$ **+ Visual Rationales**[cite: 3, 4, 5]

2.1 Stage 1: High-Recall Multi-Scorer Retrieval

An ℓ_2-normalized visual embedding $z_x \in \mathbb{R}^d$ is generated via a single pass through a frozen SigLIP-SO400M backbone (f_θ)[cite: 4]:

1. **Instance Retrieval (k-NN):** Visual embeddings of historical images are indexed using FAISS 'IndexIVFPQ' compressed to 64 bytes/vector[cite: 4, 5]. We retrieve the top $K = 50$ nearest neighbors $\mathcal{N}_K(x)$ and compute consensus voting[cite: 4]:

$$S_{\text{knn}}(c \mid x) = \frac{\sum_{i \in \mathcal{N}_K(x)} \exp(z_x^\top z_i / \tau) \cdot \mathbb{I}[c \in Y_i]}{\sum_{i \in \mathcal{N}_K(x)} \exp(z_x^\top z_i / \tau)}$$

with temperature $\tau = 0.07$[cite: 4]. Newly accepted files are appended to this index directly with zero model retraining[cite: 4].

2. **Zero-Shot Text Prototypes:** Pre-computed normalized text embeddings $t_c = g_\theta(\text{title})$ for all $|\mathcal{C}|$ categories provide zero-shot coverage for tail categories[cite: 4]:

$$s_{\text{txt}}(c \mid x) = z_x^\top t_c$$

New categories are made immediately retrievable by appending their text embeddings to prototype matrix T[cite: 4].

3. **Supervised ASL Head:** For head categories ($n \geq 30$), a linear head $p_c = \sigma(w_c^\top z_x + b_c)$ initialized with text prototypes ($w_c^{(0)} = t_c$) is trained using Asymmetric Loss[cite: 3, 4]:

$$\mathcal{L} = -\sum_c \left[y_c (1-p_c)^{\gamma_+} \log p_c + (1-y_c) p_{c,m}^{\gamma_-} \log(1-p_{c,m}) \right]$$

where $p_{c,m} = \max(p_c - m, 0)$ with $\gamma_- = 4, \gamma_+ = 0, m = 0.05$ to prevent missing labels from exerting negative gradients[cite: 4].

The top-50 candidate union C_{visual} is forwarded to Stage 2[cite: 4].

2.2 Stage 2: Metadata Discovery & DAG Pruning

Stage 2 bridges the semantic gap for non-visual categories (events, temporal spans, and administrative regions)[cite: 3]:

- **Metadata Processing:** Parses EXIF timestamps into capture years and resolves GPS coordinates into City/Country names via offline KD-tree reverse geocoding (< 2 ms)[cite: 3].
- **Proactive Event Discovery:** A fast quantized LLM ingests metadata and visual concepts to synthesize Commons event hypotheses (e.g., 'Category:[Topic] in [City] in [YYYY]')[cite: 3]. These are queried against an offline event sub-index using BM25 and text embeddings, snapping strictly to valid Commons nodes to prevent hallucinations[cite: 3, 4].
- **Conflict Filtering:** Candidates explicitly containing years or regions contradicting verified metadata are pruned[cite: 3].
- **Graph Ancestor Suppression:** We resolve cycles in the category graph using Tarjan's algorithm[cite: 3]. To follow the "most-specific" convention, an ancestor category c is suppressed if any descendant $d \in \text{descendants}(c) \cap C_{\text{pool}}$ achieves confidence $s(d) \geq 0.35$[cite: 3, 4]. If no child triggers, the ancestor is retained as a safe fallback[cite: 4].

2.3 Stage 3: Fine-Grained VLM Grounding

For the 10–15 surviving candidates, an ultra-compact VLM (Florence-2-base, 0.23B parameters) executes a single-turn visual verification pass[cite: 4, 5]. Generation is constrained via JSON schema decoding to evaluate visual presence and disambiguate sibling leaf nodes (e.g., distinguishing architectural sub-styles)[cite: 4, 5]. VLM scores are fused with prior retrieval signals[cite: 4]:

$$s_{\text{final}}(c \mid x, m) = \alpha \cdot s_{\text{prior}}(c) + \beta \cdot s_{\text{VLM}}(c \mid x)$$

Candidates lacking physical grounding are dropped, while extracted visual evidence tokens are presented to editors as human-auditable rationales[cite: 3, 4, 5].

3 Planned Testing & Evaluation Methodology

3.1 Data Splitting and De-biasing Protocol

To evaluate real-world generalization without data leakage:

- **Temporal Hold-Out Split:** Uploads are partitioned chronologically (older uploads for training/indexing, newer uploads for testing)[cite: 3, 5].
- **Near-Duplicate Deduplication:** Perceptual hashing (pHash) and visual embedding cosine thresholds (> 0.92) remove near-duplicates across splits to eliminate leakage from burst uploads[cite: 3, 6].
- **Label Cleaning:** Administrative, tracking, and licensing categories (e.g., 'Uploaded with...') are scrubbed before indexing[cite: 3, 6].

3.2 Quantitative Metrics & Stratified Evaluation

Because image labeling is multi-label, evaluation uses ranking metrics over top-k suggestions ($k \in \{5, 10\}$) against editor labels Y[cite: 3, 5]:

$$P@k = \frac{|\hat{Y}_k \cap Y|}{k}, \quad R@k = \frac{|\hat{Y}_k \cap Y|}{|Y|}, \quad F1@k = \frac{2 \cdot P@k \cdot R@k}{P@k + R@k}$$

We also record Mean Average Precision (mAP) and hierarchy-aware precision where predicted ancestors/descendants receive partial credit[cite: 3, 5]. Metrics will be stratified across:

1. **Frequency Tiers:** Head ($n > 500$), Torso ($50 \leq n \leq 500$), and Tail ($n < 50$)[cite: 3, 5].
2. **Semantic Classes:** Object, Scene, Event, and Abstract[cite: 3, 5].
3. **Image Degradation Bands:** Gaussian blur, downscaling (< 480p), and JPEG compression (quality factor ≤ 50)[cite: 3, 4, 5].

3.3 Throughput, Scalability, and Milestones

Benchmarking confirms that inference executes comfortably within the 1-second serving budget[cite: 5]:

Pipeline Component	GPU Latency
SigLIP Vision Encoder (z_x)[cite: 4]	20 ms
Stage 1: FAISS IVF-PQ ($K = 50$)[cite: 4, 5]	4 ms
Stage 1: Text Prototypes ($z_x^\top T$)[cite: 4]	2 ms
Stage 2: Metadata, Geo & Event[cite: 3, 4]	50 ms
Stage 2: DAG Ancestor Pruning[cite: 3, 4]	2 ms
Stage 3: Florence-2 VLM Verification[cite: 4, 5]	45 ms
Total End-to-End Pipeline	**123 ms**

Execution Plan: For Milestone 2 (4:00 AM), we deploy and benchmark Stage 1 retrieval alongside the DAG ancestor suppression algorithm on the temporal split to log baseline $P@k$ and $R@k$[cite: 1, 2, 4, 5]. The proactive event discovery and Stage 3 VLM reranker will be integrated for the final delivery (11:00 AM)[cite: 1, 2, 3, 4].